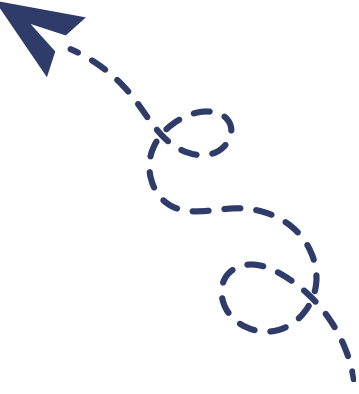




قناة تانية تمريضيانو على التليجرام



COMMUNITY HEALTH NURSING



Nursing School - Second Year



COMMUNITY HEALTH NURSING

من فضاءات التعليم الطبي والعلوم على التليجرام

Prepared By

Dr. Mayada Soliman Rashed

Ph.D. of Obstetrics and Gynecology Nursing

Dr. Hanaa El-sayed Mohamed

Ph.D. of Pediatric Nursing

Dr. Mai El-Sayed Mohsen

Ph.D. of Medical- Surgical Nursing

Dr. Abdualлах Shokrey Ismail

Ph.D. of Critical Care and Emergency Nursing

Dr. Muhammad Said Seif

Ph.D. of Medical- Surgical Nursing



In order to support the Ministry of Health and Population in the delivery of education and training for nurses, the University of Sassari and the Italian Agency for Development Cooperation are proud to share the Nursing Curriculum Model for the first level of technical education.

The curriculum, which was developed in 2022, is a benchmark for all educators who are committed to providing students with skills that are up to international standards, in order to deliver high-quality care now and in the future.

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CHAPTER ONE

(Community Health Nursing)

Learning objectives

At the end of chapter the student will be able to:-

- Define Community Health nursing and society
- Identify the concept of community
- Describe the eight characteristics of community health nursing.
- Identify Goals of community health nursing.
- Mention the role of community health nurse.

Introduction:

Community health nursing, also called public health nursing or community nursing, combines primary healthcare and nursing practice in a community setting. Community health (CH) nurses provide health services, preventive care, intervention and health education to communities or populations. Now a day's more emphasis is focused on the sick to the well person, from the individual to the community. At present public health nurses are called as Community health nurses who are registered nurses (RN) trained to work in public health settings. It includes nursing services in all phase of health services which is organized for the welfare of the community.

Community:

Community refers to a collection of people who interact with one another and whose common interests or characteristics form the basis for a sense of unity or belonging.

A society

Is a group of individuals involved in persistent social interaction, or a large social group sharing the same spatial or social territory, typically subject to the same political authority and dominant cultural expectations.

Community Health Nursing

According to the *American Nursing Association (ANA)*, Community health nursing is the practice of promoting and protecting the health of populations using knowledge from nursing, social and public health sciences

Public health

is often described as the art and science of preventing disease, prolonging life and promoting health through organized community efforts to benefit each citizen.

Community health

It refers to the health status of the members of the community, to the problems affecting their health and to the totality of the health care provided for the community.

Goals of Community Health Nursing

The primary goal of community health nursing is

- To promote health and efficiency.
- To prevent and control diseases and disabilities.
- To improve the health and well-being of individuals, families, and communities.
- To reduce health disparities and promote health equity, advocating for their communities.
- To provide direct and indirect care to individuals and families, providing support, guidance, and resources.

Principles of Community Health Nursing

- It should be planned based on the needs of the community.
- It is based on identified needs and functions.
- Integration of Health education, guidance and supervision with community health nursing services.
- Health services should be realistic in terms of available resources.
- Professional relationship with etiquette and dignity
- Individual and family member's participation in decision making.
- Health services must be continuous.
- Proper records and reports are essential.
- Proper evaluation of health services

- Health services must be available to all without any difference

Roles of Community Health Nursing

- Provide antenatal, intra-natal and postnatal, care to ensure safe pregnancy and delivery.
- administer vaccinations.
- coordinate with clinics for immunizations and wellness checks Provide under five children care.
- provide educational content and counseling
- To improve the ability of the community to deal with their own health problems
- To strengthen the community resources
- To prevent and control communicable and non-communicable diseases
- To provide specialized services
- To conduct research

Health and illness

Learning Objectives

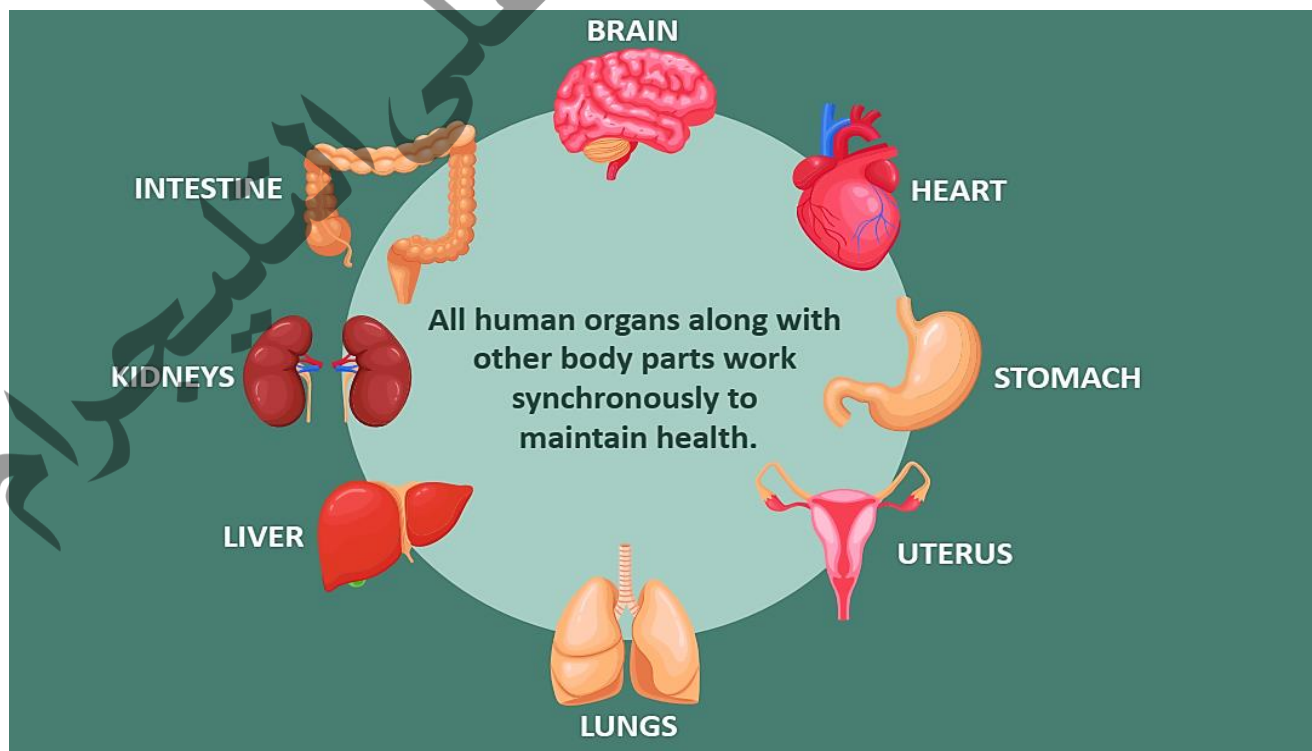
- Define health and disease.
- Discuss Determine of health and disease.
- Mention Indicators of health and disease
- Describe the basic factors to reach good health.
- Explain Prevention of disease.
- Differentiate among the three levels of prevention.

Introduction:

The human body is an amazing creation. There are many systems within the same body that function so well in a well-coordinated and integrated manner. No wonder it took some time to really understand the functioning of the human body and other related aspects. Today we have a better understanding of what constitutes health which also includes mental health. Let us take a look.

Health:

According to the **World Health Organization**, Health is defined as “the state of complete physical, mental and social well-being and not merely the absence of disease or infirmity”.



Diseases:

The term disease broadly refers to any condition that impairs the normal functioning of the body. For this reason, diseases are associated with the dysfunction of the body's normal homeostatic processes.

- Failure of any organ, organ system or body part may lead to diseases
- Sometimes microbes cause infections
- Signs & symptoms act as indicators of ruining health
- Diseases are of several types & some may lead to life-threatening & even fatal consequences in absence of appropriate medical guidance

The determinants of health include:

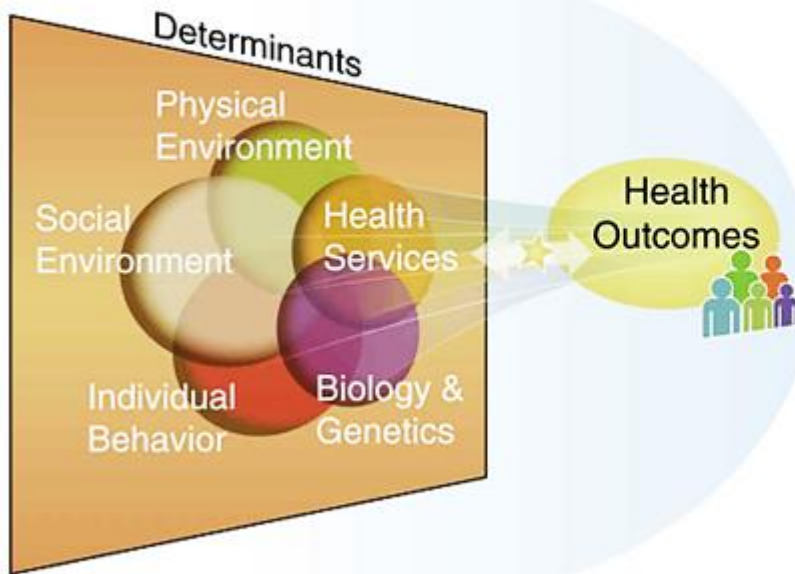
- The social and economic environment,
- The physical environment, and
- The person's individual characteristics and behaviors.

The context of people's lives determines their health, and so blaming individuals for having poor health or crediting them for good health is inappropriate. Individuals are unlikely to be able to directly control many of the determinants of health. These determinants or things that make people healthy or not include the above factors, and many others:

- **Income and social status** – higher income and social status are linked to better health. The greater the gap between the richest and poorest people, the greater the differences in health.
- **Education** – low education levels are linked with poor health, more stress and lower self-confidence.
- **Physical environment** – safe water and clean air, healthy workplaces, safe houses, communities and roads all contribute to good health.
- **Social support networks** – greater support from families, friends and communities is linked to better health.
- **Genetics** - inheritance plays a part in determining lifespan, healthiness, Personal behavior and coping skills and the likelihood of developing certain illnesses.
- **Health services** - access and use of services that prevent and treat disease influences health
- **Gender** - Men and women suffer from different types of diseases at different ages.

Healthy People 2020

A society in which all people live long, healthy lives



Overarching Goals:

- Attain high-quality, longer lives free of preventable disease, disability, injury, and premature death.
- Achieve health equity, eliminate disparities, and improve the health of all groups.
- Create social and physical environments that promote good health for all.
- Promote quality of life, healthy development and healthy behaviors across all life stages.

Model: Healthy People 2020.

From U.S. Department of Health and Human Services Office of Disease Prevention and Health Promotion: *Federal interagency workgroup: the vision, mission, and goals of healthy people 2020*, n.d., Retrieved from: <https://www.healthypeople.gov/sites/default/files/HP2020Framework.pdf>

The basic factors to reach a good health

It is well known that there are certain habits that are consistent among people who are successful, happy and productive, (and who often subsequently tend to live longer), all of which are a lot easier to adopt and practice than you might think. The following are factors that make up a healthy lifestyle.

1) Physical Activity and Fitness

Not only does exercising and staying active help keep your body healthy (especially your heart and cardiovascular system), but is also beneficial to your mental health.

2) Close Ties with Family

Forming and maintaining close and substantive relationships with your family members can contribute to your health by giving you a sense of safety and stability.

3) Maintaining a Healthy Diet

Eating foods that promote physiological well-being will simply make you feel better, which in turn will make you happy more often than depressed.

4) Getting Enough Sleep

It is proven that getting the right amount of high quality, restful sleep at the appropriate intervals has a measurable and direct impact on our health and happiness.

5) Keep Stress to a minimum

Excess stress causes chronic damage to every part of us, physically and mentally, so finding ways to avoid and alleviate stress is crucial to participating in a healthy lifestyle.

6) Sex and Intimacy

Being engaged in a healthy romantic relationship with another person is an important contributor to our happiness and mental well-being.

7) Diversify Your Hobbies

people who only watch television or surf the internet in their spare time do get sicker more often than people who have a variety of activities that they participate in such as reading, writing, painting, hiking, etc.

8) Maintain a Healthy Social Life

Of course, it's natural for some people to be introverted and others extroverted, however, being an extreme recluse and not having any kind of social life can be damaging to your happiness. Having and maintaining friendships outside of work and family is a really important part of what it means to maintain a healthy lifestyle.

9) No Tobacco

This concept is obvious for reasons we are all well aware of by now. There is no room for cigarettes or other carcinogenic tobacco products in a healthy lifestyle regimen.

Disease Prevention

Infectious diseases are caused by microscopic organisms that live in other people, animals, or the environment and are too small to see. If you do not come into contact with them, you can prevent many infections and illnesses.

Prevention includes a wide range of activities known as “interventions” aimed at reducing risks or threats to health

Categories (levels) of prevention

Primary prevention

Primary prevention aims to prevent disease or injury before it ever occurs. This is done by preventing exposures to hazards that cause disease or injury, altering

unhealthy or unsafe behaviors that can lead to disease or injury, and increasing resistance to disease or injury should exposure occur. **Examples include:**

- education about healthy and safe habits (e.g. eating well, exercising regularly, not smoking)
- immunization against infectious diseases.

Secondary prevention

Secondary prevention aims to reduce the impact of a disease or injury that has already occurred. This is done by detecting and treating disease or injury as soon as possible to slow its progress, encouraging personal strategies to prevent reinjury or recurrence, and implementing programs to return people to their original health and function to prevent long-term problems. **Examples include:**

- regular screening tests to detect disease in its earliest stages (e.g. mammograms to detect breast cancer)
- daily, low-dose aspirins and/or diet and exercise programs to prevent further heart attacks or strokes

Tertiary prevention

Tertiary prevention aims to soften the impact of an ongoing illness or injury that has lasting effects. This is done by helping people manage long-term, often-complex health problems and injuries (e.g. chronic diseases, permanent impairments) in order to improve as much as possible their ability to function, their quality of life and their life expectancy. **Examples include:**

- Cardiac or stroke rehabilitation programs, chronic disease management programs (e.g. for diabetes, arthritis, depression, etc.)
- support groups that allow members to share strategies for living well
- vocational rehabilitation programs to retrain workers for new jobs when they have recovered as much as possible.

Principles of infection Prevention

1. Vaccinate & Use Medicines Properly

- Keep immunizations up to date. Follow recommended immunizations for children, adults, and pets.

- Use antibiotics exactly as prescribed. Take them for the full course prescribed by your doctor, Never self-medicate with antibiotics or share them with family or friends.
- Report to your doctor any quickly worsening infection or any infection that does not get better after you take a prescribed antibiotic.
- If you travel internationally, get all recommended immunizations, and use protective medications for travel, especially to areas with malaria.

2. Keep Clean

- Wash your hands often, especially during cold and flu season.
- Be aware of what you eat, and prepare foods carefully.

3. Protect Yourself from Disease Carriers

- Be cautious around all wild and domestic animals that are not familiar to you.
- After any animal bite, clean the skin with soap and water, and seek medical care immediately.
- Avoid areas where there are ticks.
- Protect yourself from mosquitoes.

4. Don't Spread Disease

- If you are sick with a cold or flu, stay home and don't spread germs.
- Do not use intravenous drugs or share syringes.

Healthcare team

Learning Objectives

- Define healthcare team.
- Mention Healthcare team member.
- Discuss the Role of healthcare team member.
- Describe the Factors that raise the efficiency of the health team.

Definition of healthcare team

It is the group of professionals who contribute to patient care and treatment. there might be a Nurse Practitioner instead of or as well as a doctor. In others, physical therapists, occupational therapists, and social workers may be part of the team.

Healthcare team member

Healthcare is a team effort. Each healthcare provider is like a member of the team with a special role. Some team members are doctors or technicians who help diagnose disease. Others are experts who treat disease or care for patients' physical and emotional needs.

Healthcare team members include:

- Physician
- Registered nurse
- Pathologist
- Pharmacist
- Radiologist
- Dietician
- Social worker
- Spiritual leader

Role of healthcare team member: -

➤ Physicians:

Doctors or physicians are key members of the healthcare team. They invest years in education and training. They may be primary care doctors or specialist

- **Primary care doctors:** Primary care doctors mainly focus on preventive care. It includes routine check-up, immunization and counselling.
- **Specialist:** Specialist doctors got training in special areas of health. For example, *neurologists, endocrinologist, Ortho, pediatricists*.

➤ **Nurse:**

They provide care, administer medication and coordinate with other healthcare team members. They have got training around 3-4 years depending on the advancement of course. They are highly skilled professionals who provide excellent care to the patient.

➤ **Pathologist:**

Pathologist perform examinations of body fluid and tissues in the laboratory. They help in reaching a particular diagnosis of patient.

➤ **Pharmacist:**

Pharmacist is responsible for checking quality, safety and dispensing of medications. They are also responsible for storing and maintaining stock of drug in the healthcare setting.

➤ **Radiologist:**

Radiologists are specialized in imaging techniques like CT scan, MRIs. They use a wide range of tools to get the image of a particular organ.

➤ **Dietician:**

Dietician is responsible for determining the nutritional needs, observing preparation of food, checking quality and preparing diet plan for the patient.

➤ **Social Worker:**

They provide suggestions regarding home care, educating the patient about disease conditions and facilities. Most healthcare social workers involved in mental illness.

➤ **Spiritual Leader:**

Spiritual leader is also included in healthcare team to meet spiritual needs of the patient. Every member of the healthcare team plays their role in their particular area of expertise.

Factors that raise the efficiency of the health team.

➤ Overcoming Barriers to Teamwork

Building better healthcare teams is, in part, about overcoming the educational, psychological and organizational factors that frustrate our efforts to collaborate with each other. How do we do this? Here are seven proven strategies.

➤ Teaching Effective Communication Strategies

Teaching structured methods of communication, such as 'SBAR' handovers, can improve patient outcomes.

➤ Train Teams Together

Teams that work together should train together, enabling them to better understand each other's roles and contributions.

➤ Train Teams Using Simulation

Using simulation in a safe way to practice new techniques and rehearse frequently encountered situations improves collaboration and team effectiveness.

➤ Define Inclusive Teams

Redefine your team. Rather than seeing yourselves as a collection of disciplines, create common and cohesive goals that all members share.

CHAPTER TWO

Part (1): Primary Health Care

Learning objectives

At the end of chapter the student will be able to:-

- Define primary health care (PHC).
- Recognize the component of PHC.
- Identify the important of PHC.
- Explain principles of PHC.
- Classify types of PHC team.
- Discuss the role of nurse in PHC.

Introduction:

All people, everywhere, deserve the right care, right in their community. This is the fundamental premise of primary health care. Primary health care addresses the majority of a person's health needs throughout their lifetime. This includes physical, mental and social well-being and it is people-centered rather than disease-centered.

Definition of primary health care:

Primary health care (PHC) is a whole-of-society approach that includes health promotion, disease prevention, treatment, rehabilitation and palliative care.



The components of primary health care:

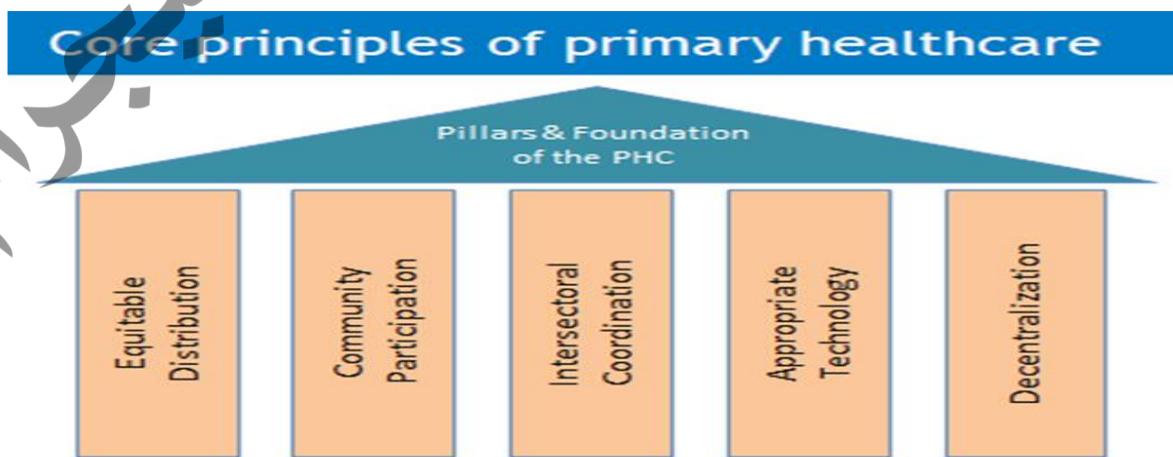
1. Meeting people's health needs through comprehensive promotive, protective, preventive, curative, rehabilitative, and palliative care throughout the life course;
2. Systematically addressing the broader determinants of health (including social, economic and environmental factors) through evidence-informed policies and actions across all sectors; and
3. Empowering individuals, families, and communities to optimize their health.



Important of PHC:

- PHC is the most inclusive, equitable, cost-effective and efficient approach to enhance people's physical and mental health, as well as social well-being .
- PHC is also critical to make health systems more resilient to situations of crisis, more proactive in detecting early signs of epidemics and more prepared to act early such as COVID-19.

Principles of PHC:



1- Equitable Distribution of Resources:

All human beings have an equal right to health. Equitable distribution means that health services must be shared equally by all people irrespective of their ability to pay; and all the people -rich or poor, rural or urban –must have access to health services.

2- Community participation:

Community participation is the process by which individuals, families and communities assume responsibility in promoting their own health and welfare. The PHC strategy underlines the importance of full community participation, especially in health decision-making.

3- Inter-sectoral Coordination:

PHC requires the support of other sectors such as sectors responsible for economic development, water purification, sanitation, communication, environmental protection and education all contribute to health.

4- Appropriate healthcare technologies:

Appropriate technology refers to using cheaper, scientifically valid and acceptable equipment and techniques.

- It is also necessary to ensure that the technology is:
- Scientifically reliable and valid.
- Adapted to local needs.
- Acceptable to the community people.
- Accessible and affordable by the local resources.

5- Decentralization:

PHC promotes people centered approach to healthcare and this encompasses promotive, preventative, curative and rehabilitative services through community collaboration and full participation.

Elements of Primary Health Care (PHC):

There are eight (8) elements of Primary Health Care. These 8 elements are also known as ‘essential health care’.

Elements of Primary Health Care (PHC)	
E	Education about prevailing health problems and methods of preventing and controlling them
L	Prevention and control of locally endemic diseases
E	Provision of essential drugs
M	Maternal and child health care; including family planning
E	Expanded immunization against major infectious diseases
N	Promotion of food supply and proper nutrition
T	Appropriate treatment of common diseases and injuries
S	Adequate supply of safe water and basic sanitation

Health Care Team in PHC:

Different types of teams can be identified in health care systems:

- **Core teams:** These are directly involved in caring for the patient.
 - They usually consist of team leaders and members who are direct-care providers such as nurses, dentists, pharmacists, doctors, assistants...etc.
- **Coordinating teams:** The group responsible for coordinating functions and resource management for core teams.
- **Contingency teams:** Formed to deal with emergencies or specific events (e.g. cardiac-arrest teams, disaster-response teams...etc).
- **Ancillary teams/services:** The group supports services that facilitate patient care such as cleaners or domestic staff.
- **Support services and administration:** Those who provide indirect, task-specific services in a health-care facility support service. It includes secretaries and the executive leadership of a unit or facility. This team has 24-hour accountability for the overall functioning of the organization.

Role of nurse in primary health care:

Nurses working in primary health care should play the following roles

1- Patient advocacy and education:

Ability to promote patients' entitlement to ensure the best quality of care and empowering patients to become active participants of their health.

2- Effective Communication:

Ability to quickly establish rapport with patients and their family members.

3- Team-work and leadership:

Ability to function effectively as a member of an interprofessional team that includes providers, patients and family members to building productive working relationships and is focused on health outcomes.

4- People-centered care and clinical practice:

Ability to create conditions for providing coordinated/integrated services centered on the patients and their families' needs, values and preferences.

5- Continuous learning and research:

Ability to demonstrate reflective practice, based on the best available evidence and to assess and continually improve the services delivered.

Part (2): Maternal and child health

Learning objectives

At the end of chapter the student will be able to:-

- Define maternal health, child health (childhood classification).
- Recognize the relation between maternal health and child health.
- Identify maternal and child health care (MCH) services.
- List objectives of MCH services.
- Recognize Major component of MCH service.
- Explain Common indicator of MCH Service.
- Discuss partnership to improve MCH in Egypt.
- Demonstrate the Role of Nurses in MCH Care.

Introduction:

A person's health at each stage of life affects health at other stages and also can have cumulative effects for the next generation. Women who remain healthy during pregnancy and after birth are more likely to stay healthy later in life and have better birth outcomes, influencing infancy, childhood and adulthood.

Maternal and child health:

Maternal and child health (MCH) refers to the health of mothers, infants, children, and adolescents. It also refers to a profession within public health committed to promoting the health status and future challenges of this vulnerable population.

Definition of maternal health:

Maternal health refers to the health of women before and during pregnancy, at childbirth and during the postpartum period.

Definition of child health:

Children's health is the extent to which individual children or groups of children are able or enabled to (a) develop and realize their potential, (b) satisfy their needs, and (c) develop the capacities that allow them to interact successfully with their biological, physical, and social environments.

» Developmental stages of childhood:



Early childhood:

Early childhood follows the infancy stage and begins with toddlerhood when the child begins speaking or taking steps independently, early childhood continues approximately until the age of 6 or 7.

Middle childhood:

Middle childhood begins at around age 7, approximating primary school age. It ends with puberty (around age 12 or 13), which typically marks the beginning of adolescence.

Post-childhood (Adolescence):

Adolescence is usually determined to be between the onset of puberty and legal adulthood: mostly corresponding to the teenage years (13-19). However, puberty usually begins before the teenage years.

Relation between maternal health and child health:

Poor maternal health and inadequate maternal nutrition are two of the major determinants of child health. Main effects include **low birth weight, overweight/underweight, anemia, infections, immune disorders and increased child's risk of obesity and related chronic diseases as an adult.**



Maternal and child health services:

MCH services defined as promoting, preventing, therapeutic or rehabilitation facility or care for the mother and child.

▪ Objectives of MCH services:

- 1-Reduction of maternal, perinatal, infant, and childhood mortality and morbidity
- 2-Promotion of reproductive health
- 3-Promotion of physical and psychological development of the child and adolescent within the family

The ultimate objective of MCH service is life long-health.

Component of maternal and child health (MCH) services in Egypt:

- 1- Prenatal, postnatal care.
- 2- Child health care and immunization
- 3- Family planning
- 4- Reproductive health

MCH programs focus on health issues concerning women, children and families, such as access to recommended prenatal and well-child care, infant and maternal mortality prevention, maternal and child mental health, newborn screening, child immunizations, child nutrition and services for children with special health care needs.

Maternal and child health indicators:

Maternal Health Indicators:

1- Maternal mortality ratio

Maternal mortality ratio (MMR) is the number of women who die from pregnancy-related causes while pregnant or within 42 days of pregnancy termination per 100,000 live births. **Most maternal deaths result from:** Postpartum hemorrhage, hypertensive diseases, antepartum hemorrhage, and sepsis.

2- Antenatal care coverage:

Many health problems in pregnant women can be prevented, detected and treated during antenatal care visits with trained health workers.

3- Skilled birth deliveries:

This indicator is directly linked to a process of giving birth by pregnant woman, and most of maternal mortality is likely to happen at this stage. Medically assisted deliveries dramatically increased.

4- Postnatal care coverage:

Since up to 50 percent of maternal deaths occur after delivery, a midwife or a trained and supervised Traditional Birth Attendant (TBA) should visit all mothers as soon as possible within the first 24–48 hours after birth.

5- Use of modern family planning methods:

The 2014, Egypt Demographic and Health Survey (EDHS) indicate that around 59 percent of currently married women in Egypt are using contraception. The most widely used methods are the IUD, the pill, and injectable.

Child Health Indicators:

1- Infant and Under-five mortality rate:

Under-five mortality rate is the probability per 1,000 that a newborn baby will die before reaching age five.

2- Vaccination coverage:

The children should receive complete vaccination against tuberculosis, measles, diphtheria, pertussis, tetanus and polio.

3- Exclusive breastfeeding:

The global public health recommendation is that infants should be exclusively breastfed for the first six months of life, starting in the first half hour after delivery. In Egypt, the EDHS, 2014 shows that exclusive breastfeeding is common but not universal in very early infancy.

4- Child malnutrition:

With two thirds of child mortality attributable to malnutrition. Malnutrition rates remain high particularly among children under-five.

Partnership to improve maternal and child health in Egypt:

Egypt made positive advances in maternal and child health over the past twenty years, Egypt is close to meeting its sustainable Development Goal (SDG 3): Ensure healthy lives and promote well-being for all at all ages.

➤ Targets:

- By 2030, reduce the global maternal mortality ratio to less than 70 per 100,000 live births.
- By 2030, ensure universal access to sexual and reproductive health-care services, including for family planning, information and education, and the integration of reproductive health into national strategies and programs.
- By 2030, end preventable deaths of newborns and children under 5 years of age, with all countries aiming to reduce neonatal mortality to at least as low as 12 per 1,000 live births and under-5 mortality to at least as low as 25 per 1,000 live births.

Role of nurse in MCH care:

▪ Service Provider

- Provision of antenatal care.
- Monitoring the growth of the fetus & its wellbeing.
- Screening for mal formations, congenital anomalies & other deviations.
- Supplementation of requisite vitamins & micro nutrients.
- Promotion of good delivery practices.
- Promotion of breast feeding & maternal bonding.
- Promotion of optimal new born care
- Ensuring appropriate immunization services.
- Periodic growth, development & mile stone monitoring.

▪ Services for Mothers

- Assess for feeding difficulties.
- Promote iron & calcium supplementation.
- Monitor & promote balanced diet & good dietary habits.
- Monitor for post natal blues.
- Promote adoption of contraception & family planning practices.
- Educate the mother on the importance of child rearing & self-care practices including diet, exercise & sleep.

▪ **Role as an Administrator:**

- Serve as an information provider to policymakers in relation to MCH services.
- Serve as a liaison with the government, community & NGO in organizing & implementing MCH services.
- Develop counseling & MCH assistance network.

▪ **Educationist**

- Organize training programs for health care professionals.
- Conduct workshops & conferences relating to MCH care & sensitize the community, and health care professionals.
- Design health education materials about MCH services.

▪ **Researcher**

- Identify researchable areas in MCH & conduct research.
- Coordinate & network on research activities.
- Design models based on research findings.

Students' self-assessment

A. which team is responsible for a dealing with emergencies or specific events

- a. Core teams
- b. Coordinating teams
- c. Contingency teams
- d. Ancillary teams/services

B. All the following represent Child Health Indicators, except:

- a. Infant and Under-five mortality rate
- b. Vaccination coverage
- c. Skilled birth deliveries.
- d. Exclusive breastfeeding

C. Most maternal deaths result from:

- a. Postpartum hemorrhage.
- b. Diabetes mellitus.
- c. Low birth weight.
- d. Congenital anomalies.

D. Discuss the role of community nurse in MCH care.

CHAPTER THREE

PUBLIC HEALTH SERVICES

Learning objectives

At the end of chapter the student will be able to:-

1. Define community health services
2. Discuss home health care
3. Explain family health care
4. Describe school health care

Introduction:

Public health services cover a wide range of services and provide care for people from birth to the end of their life. Community health teams play a vital role in supporting people with complex health and care needs to live independently in their own home for as long as possible. Services also include health promotion services such as school health services and health visiting services.

Definition of public health services

Services are delivered in a wide range of settings – including in people's own homes as well as in community clinics, community centers and schools.

Universal public health functions

- Health visiting
- School nursing
- Family & Sexual health services

Home health care

Introduction

Community Health at Home provides treatment and assistance to patients who are recuperating, chronically ill, or disabled - in the comfort of their own home. Our home care services include specialty programs, prevention and health maintenance on a short-term and long-term basis, and a wide array of services from high-technology nursing to basic home health aide care.

Home Health Care Defined

➤ Four Different Perspectives

▪ Official

Services are provided to individuals and their families in their place of residence for the purpose of promoting, maintaining, or restoring health or of minimizing the effects of illness and disability.

▪ Patient

Skilled and compassionate care is provided on a one-to-one basis in the comforting and familiar surroundings of the home.

▪ Family

It is a means to keep the family together as a functioning, integrated unit.

▪ Provider

All disciplines involved are challenged to provide excellent care in often less-than-excellent conditions and surroundings.

Home Health Nursing

- Home health nursing services enable individuals of all ages to remain in the comfort and security of their homes while receiving health care.

Home health services components:

- Services may include skilled nursing, physical therapy, speech language therapy, occupation therapy, social services, intravenous therapy, nutritional support, home health aide, respiratory therapy, acquisition of medical supplies and equipment, and homemaker and companion care.



Skilled Nursing:

Skilled Registered Nurses have the experience and resources to provide a number of services, including:

- Infusion Therapy & Injections
- Lab Draws
- Central Line & Port Maintenance
- Wound Care
- Tube Feeding
- Ostomy Care
- Treatment for Skin Conditions
- Disease Management

Physical Therapy:

There are specific healing techniques that only a trained physical therapist knows. Most patients prefer to do physical therapy sessions in the privacy of their own home. With a physical therapist, patients will see many benefits, including:

- Improved endurance
- Rehabilitation following surgery
- Enhanced gait (walking technique) training
- Increased mobility and strength

Occupational Therapy:

An occupational therapist is an ideal support benefit for a new caregiver or family member who may not know what their patient needs. For example, there may be safety hazards that an occupational therapist can help identify. Having an occupational therapist come to your home can help in these ways:

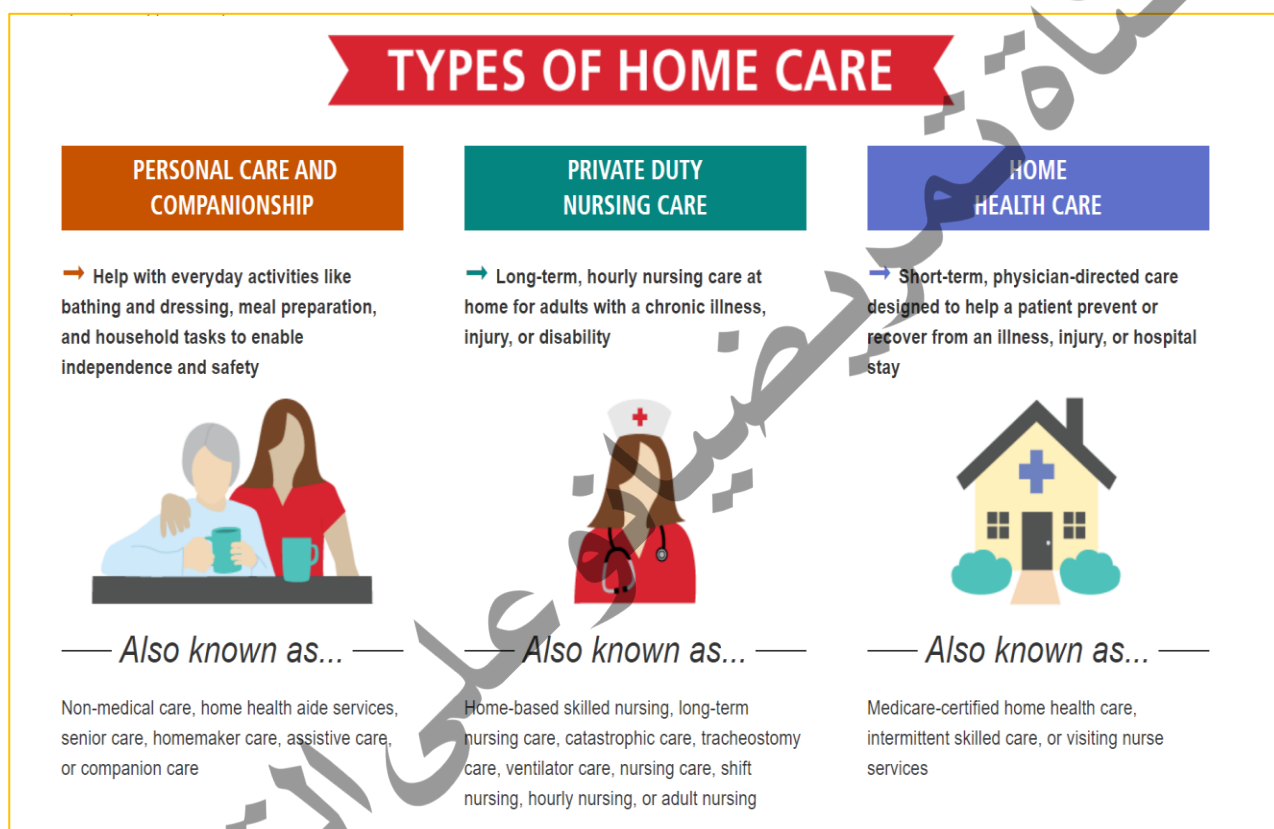
- Improve patient's ability to perform daily self-care
- Restore independence in the home
- Provide adaptive equipment recommendations for the home

Speech Therapy

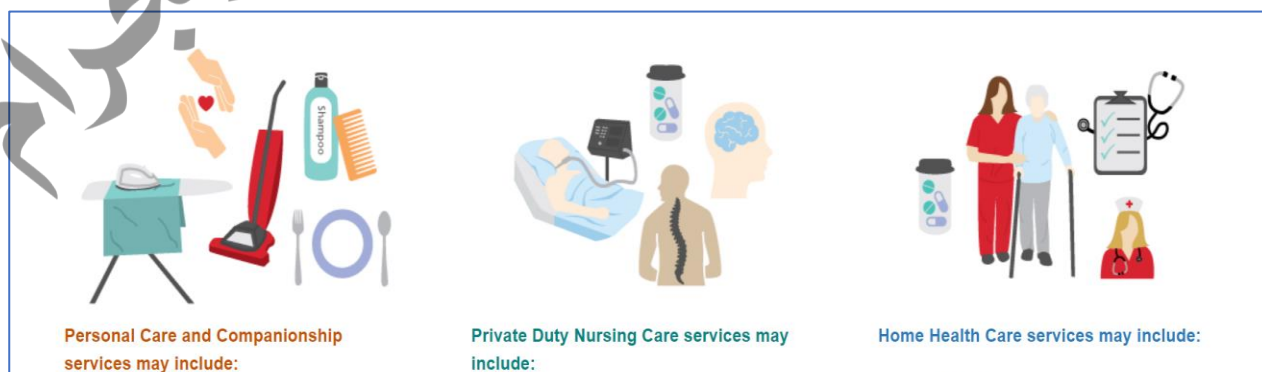
A speech therapist is important for regaining the ability to speak and improving cognitive deficiencies. Speech therapy is often needed after a stroke. With regular sessions, patients will see:

- Improvement in ability to speak
- Improvement in swallowing
- Improvement in cognitive deficiencies

Types of home care:



Examples of care services



Personal Care and Companionship services may include:

- Assistance with self-care, such as grooming, bathing, dressing, and using toilet.
- Enabling safety at home by assisting with ambulation, transfer (eg, from bed to wheelchair, wheelchair to toilet), and fall prevention
- Assistance with meal planning and preparation, light housekeeping, laundry, errands, medication reminders, and escorting to appointments
- Companionship and engaging in hobbies and activities
- Supervision for someone with dementia or Alzheimer's disease.

Private Duty Nursing Care services may include:

- Care for diseases such as Traumatic brain injury or Spinal cord injury.
- Ventilator care
- Tracheostomy care
- Monitoring vital signs
- Administering medications
- Ostomy/gastrostomy care
- Feeding tube care
- Catheter care

Home Health Care services may include:

- Short-term nursing services
- Physical therapy
- Occupational therapy
- Speech language pathology
- Medical social work
- Home health aide services

Home visit

Introduction:

A home visit is one of the essential parts of the community health services because most of the people are found in a home. Home visit fulfils the needs of individual, family and community in general for nursing service and health counselling. A home visit is considered as the backbone of community health service.

A home visit is a family–nurse contact which allows the health worker to assess the home and family situation in order to provide the necessary nursing care and health-related activities. The services in the home requires technical skills, knowledge of preventive and therapeutic measures, teaching ability, judgment and full understanding of human relation.

Definition:

- A home visit is defined as the process of providing the nursing care to patients at their doorsteps. It requires technical skills, resourcefulness, judgment, relationships.
- It is defined as providing the services to family at their door step to maintain the health & to reduce the mortality & morbidity in family.

Purposes of home visiting:

1. To give care to the sick, to a postpartum mother and her newborn with the view teach a responsible family member to give the subsequent care.
2. To assess the living condition of the patient and his family and their health practices in order to provide the appropriate health teaching.
3. To give health teachings regarding the prevention and control of diseases.
4. To establish close relationship between the health agencies and the public for the promotion of health.
5. To make use of the inter-referral system and to promote the utilization of community services.

Principles of home visiting

1. Home visit must have purpose or objective.
2. Planning for a home visit should make use of all available information about the patient and his family through family records.
3. Consider and give priority to the essential needs of the individual and his family
4. Planning and delivery of care should involve the individual and his family.
5. The plan should be flexible.

Guidelines to consider regarding the frequency of home visits

- **Need Based:** home visiting should be planned and conducted based on the identified needs of the people
- **Priority based:** the home visit should give to the existing problem in the family. It may be maternal and child health services or antenatal checkup
- **Regularity:** plan for regular home visiting programs based on family needs. It should be conducted at regular intervals
- **Flexibility:** the community health nurse should adopt a flexible approach based on prevailing circumstances at home
- **Scientific based:** be sure of the scientific soundness of the subjects used for discussion. Use of technical skills includes hand washing and inspection
- **Analysis based:** collect facts about the home, the patient and the environment and make an objective analysis of the facts as an initial step in visiting the home
- **Developing relationship:** work with the person and family plan jointly. Home visiting helps to establish good working relationship in the family
- **Sensitivity:** the community health nurse should be sensitive to the persons feeling and needs at the time of the visit. Listen to the family and understand the other person's point of view
- **Evaluation:** evaluate the quality of care is more important than the number of home visits. It is essential to evaluate home visits from time to time

Phases of home visiting:

1. **Initial phase:** the community health nurse should
 - Collect information from clinical and other records before planning for a visit.
 - Assess or observe and make a note in initial visit

- Introduce and establish a friendly relationship by using simple language.
- Assess physical and environmental status, family's cultural background, occupation and income of family members, age, educational factors and psychological factors influences

Action phase: *The nurse should*

- Use their effective communication skills to implement the nursing process.
- Practice a variety of roles when interviewing in patient care.
- Take a role as collaborator, consultant, co-coordinator, preventor of disease, promoter of health, health educator and an epidemiologist, and takes steps to implement nursing process
- Provides nursing care, e.g. taking temperature, physical examination and dressing, etc. demonstrating and teaching, e.g. teaching insulin self-administration. She makes diagnosis and tentative nursing care plan based on establishing priorities

Termination phase:

- Nurse-patient goals are reached; health is restored and the patient can function without actions.
- The nurse records the important events in the family and reports the problems of the family. Evaluation of home visit is a continuous process.

Steps

1. Greet the patient and introduce yourself.
2. State the purpose of the visit
3. Observe the patient and determine the health needs.
4. Put the bag in a convenient place and then proceed to perform the bag technique.
5. Perform the nursing care needed and give health teachings.
6. Record all important date, observation and care rendered.
7. Make appointment for a return visit.

Advantages of home visit

- The nurse can directly observe home and family atmosphere
- The nurse can directly observe the care given to patient by the family members
- It is possible to discover new health problems
- The family members will be more relaxed in their own surroundings
- The family gains confidence and feels to clear their doubts
- This helps to apply the gained knowledge and skills in the homes assisting and solving individuals and family's health problems.

Disadvantages of home visits:

- Home visits are time consuming; travel time is required to get to the persons needing care.
- There is no easy access to emergency equipment or consultation with other health professionals if needed.
- Home visits may present issues regarding the nurse personal safety in some community or family settings.
- The nurse has less control over the care setting (for example, cleanliness, noise, privacy, or distractions).

Community health nurse bag:

The **public health bag** is an essential and indispensable equipment of a public health nurse which she has to carry along during her home visits. It contains basic medication and articles which are necessary for giving care.

The **bag technique** is a tool by which the nurse, during her visit will enable her to perform a nursing procedure with ease and deftness, to save time and effort with the end view of rendering effective nursing care to clients.

Principles

- Performing the bag technique will minimize, if not, prevent the spread of any infection.
- It saves time and effort in the performance of nursing procedures.

- The bag technique can be performed in a variety of ways depending on the agency's policy, the home situation, or as long as principles of avoiding transfer of infection is always observed.

Contents

The following are the contents of a Public Health Nurse bag:

- Paper lining/ Extra paper for making waste bag
- Plastic/linen lining
- Apron/Hand towel
- Soap in a soap dish
- Thermometers (oral and rectal)
- 2 pairs of scissors (surgical and bandage)
- 2 pairs of forceps (curved and straight)
- Disposable syringes with needles
- Hypodermic needles
- Sterile dressing / Cotton balls
- Cord clamp
- Micropore plaster/Tape measure

- 1 pair of sterile gloves
 - Baby's scale
 - Alcohol lamp
 - 2 test tubes/ Test tube holders
 - Solutions of:
 - Betadine, 70% alcohol, Zephiran solution, Hydrogen peroxide
 - Ophthalmic ointment
 - Acetic acid
 - Benedict's solution
- *BP apparatus and stethoscope are carried separately and are never placed in the bag.**

Points to consider:

1. The bag should contain all the necessary articles, supplies and equipment that will be used to answer the emergency needs
2. The bag and its contents should be cleaned very often, the supplies replaced and ready for use anytime.
3. The bag and its contents should be well protected from contact with any article in the patient's home.
4. Consider the bag and its contents clean and sterile, while articles that belong to the patients as dirty and contaminated.
5. The arrangement of the contents of the bag should be the one most convenient to the user, to facilitate efficiency and avoid confusion.

Family Health Care Services

Introduction:

Family health nursing care services in the community are the focal point of health services. It is special care given to family members to promote their health, prevent health problems, and for the well-being of the family.

Definition:

Family: the basic unit in society, is two or more persons related by birth, marriage, or adoption who reside together in a household, and is made up of all the forces that surround it. They constitute a single home, interact with each other in their respective family roles, and create and maintain a common culture.

"Family health services can be defined as having the skills and resources to carry out family development tasks. It is special care given to family members to promote their health, prevent health problems, and for the well-being of the family."

Family nursing: is part of the primary care provided to patients of all ages, from child health to geriatric. This helps foster a strong relationship between the healthcare provider and the patient. Family nursing is not a care centered so much on the patient as on the care of the family unit.

Purpose:

1. Reduce maternal mortality, morbidity and the infant mortality rate.
2. Spacing of children's births.
3. Solve the problem of malnutrition in the family.
4. Health education

Scopes:

The goals of individual and community health and growth at the national health level can be achieved through family health services:

1. Reproductive health
2. Children's health
3. Adolescent health

4. Mental health
5. Gender issues
6. Ageing

Principles:

- Establish a good professional relationship with the family
- Knowledge of the basic facts Size, occupation, such as norms of customs, rituals and education
- Identification and priority of problems
- Need-based services and supports for the family
- Discussion of the problem with the family and possible solution
- Encourage the family to be self-sufficient to meet their needs
- Health services must be independent of gender, age, income and religion
- Involvement of family members

Characteristics of Healthy Family:

1. A facilitative process of interaction exists among family members.
2. Individual member development is enhanced.
3. Role relationships are structured effectively.
4. Active attempts are made to cope with problems.
5. There is a healthy home environment and lifestyle.
6. Regular links with the broader community are established.

Family Health Assessment

Assessing family health in a systematic fashion require three tools

1. Conceptual framework
2. Data collection categories
3. Assessment methods

1. Conceptual frameworks

A conceptual framework is a set of concepts integrated into meaningful explanation that helps one interpret human behavior or situations. Three frameworks that are particularly useful in community health nursing are presented here:

1. Interactional frameworks:

Describes the family as a unit of interacting personalities and emphasizes communication, roles, conflict, coping patterns, and decision-making processes

2. Structural-functional frameworks:

Describes the family as a social system relating to other social systems in the external environment, such as school, work, and the health care system.

3. Developmental frameworks:

Studies families from a life-cycle perspective by examining members' changing roles and tasks in each progressive life-cycle stage

2. Data collection categories

1. Types and kinds of data needed

1. Family structure, characteristics, and dynamics
2. Socio-economic and cultural characteristics
3. Home and environment
4. Health Status of each member
5. Values and practices on health promotion and maintenance and illness prevention.

3. Method of data gathering

- primary source of data collection- obtained directly from the client (family members)
- Secondary source of data collection- obtained through friends, neighbors, colleagues, family records, family team members, investigation reports, reference books etc.

4. Assessment methods

- Many different methods are used to assess families.
- These methods serve to generate information about selected aspects of family structure and function; the methods must match the purpose for assessment.

1. Ecomap

Is diagram of the connection between a family and the other system in its ecological environment or its A picture of the family's patterns.

2. Genogram

A graphic picture of family history, usually used over three or more generation. The genogram maps such information as/; a. Relationships among family members

3. Social Network Support Map or Grid

Gives details about the quality and quantity of social connections.

4. Family Health Tree a record of diseases that occur in a family.

Family health nursing process:

It is a systematic, problem-solving, logical, and deliberate process to help the family develop and strengthen their capacities to meet their health needs and solve their health problems. It consists of five phases:

1. Assessment

Involves Observation, Nursing History, Literature Review Consultation, Interview, and Clinical History, Physical and Psychosocial History.

2. Diagnosis:

It consists of the family diagnosis.

3. Planning:

Involves setting priorities, setting goals and objectives, and formulating a family health nursing care plan.

4. Implementation:

It consists of the mobilization of resources facilitating the work environment, implementation and documentation.

5. Evaluation:

It involves the summative and formative evaluation of the family health nursing care plan.

Family crisis

Definition of a Crisis:

A disruption or breakdown in a person's or family's normal or usual pattern of functioning. A crisis cannot be resolved by a person's customary problem-solving resources/skills.

Three basic elements of a crisis are:

A stressful situation, difficulty in coping, and the timing of intervention. Each crisis situation is unique and will require a flexible approach to the client and situation.

Situations Which Can Lead to a Crisis

- **Family Situations** - a child abuse investigation, spouse abuse, an unplanned pregnancy, a parent's desertion, a chronically ill family member, and lack of social supports are examples of family situations that can create stress and crises.
- **Economic Situations** - sudden or chronic financial strain is responsible for many family crises, such as loss of employment, eviction, no food, a theft of household cash or belongings, high medical expenses, missed child support payments, repossession of a car, utilities cut off from service, money "lost" to gambling or drug addiction, and poverty.
- **Community Situations** - neighborhood violence, inadequate housing, a lack of community resources, and inadequate educational programs illustrate some ways the community may contribute to family crises.
- **Significant Life Events** - events that most view as happy, such as a marriage, the birth of a child, a job promotion, or retirement, can trigger a crisis in a family; a child enrolling in school, the behaviors of an adolescent, a grown child leaving the home, the onset of menopause, or the death of a loved one can also be very stressful life events.
- **Natural Elements** -crises are created by disasters such as floods, hurricanes, fires, and earth quakes, or even extended periods of high heat and humidity, or gloomy or excessively cold weather.

Intervention of Crisis:

☒ Case workers and disability program specialists respond by doing the following:

- Take a quick inventory of the situation
- Identify the type of crisis
- Take action to resolve the situation.
- Attempt to defuse situation and/or reassure the client;

☒ Once the situation is calm:

- Identify and contact available community resources in your area that can assist the client through the crisis
- Document events to the extent possible, maintaining confidentiality when required.

☒ Maintain professional skills and resources:

- Identify and post information about available community resources. Keep that information available to all clients and staff.
- Seek additional training opportunities when available.

Referral system

Definition of referral system

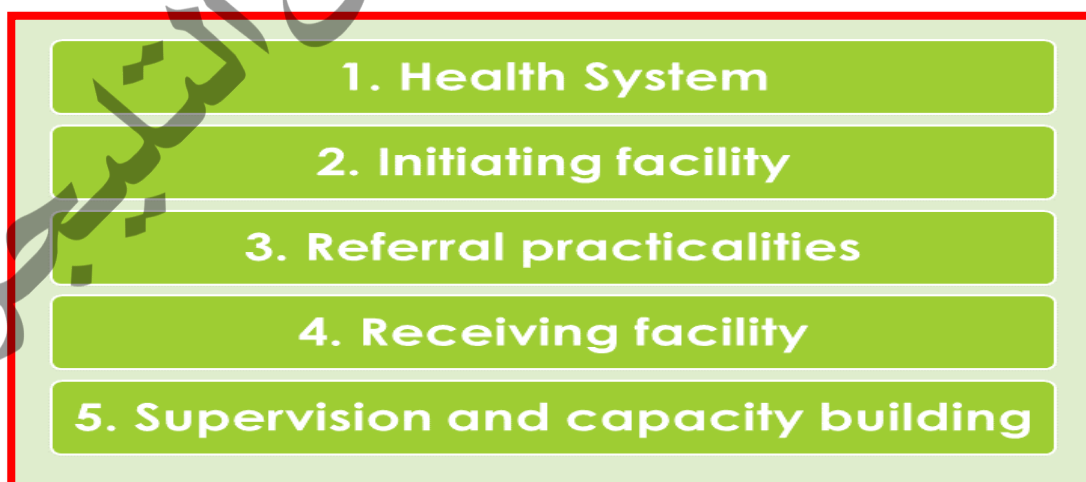
"Referral system is defined as a system of transferring cases which are beyond the technical competence of one infrastructure to higher level infrastructure / institution having technical competency and all other resources to provide desired health services".

In another word "a referral can be defined as a process in which a health worker at a one level of the health system, having insufficient resources (drugs, skills, equipment) to manage a clinical condition, seeks the assistance of a better or differently resourced facility at the same or higher level to assist in, or take over the management of, the client's case.

Purpose: A good referral system can help to ensure:

- Clients receive optimal care at the appropriate level and not unnecessarily costly.
- Hospital facilities are used optimally and cost-effectively.
- Clients who most need specialist services can accessing them in a timely way.
- Primary health services are well utilized and their reputation is enhanced.

Components of a referral system



Types of referral system:

Types of Referral		
Routine	Emergency	Opportunistic
• Seeking expert opinion for diagnosis and prognosis. • Seeking hospital admission and management for the case. • Seeking further investigations	• To reach the expert on time before occurrence of deterioration with providing all expected information in referral letter.	• For expert opinion • Investigation • Admission • Management

Role of the Family Health Nurse

- Collaborate between family and health professionals
- Function as client advocate
- Promote and provide health education
- Assist family to access resources
- Guide families to identify strengths
- Establish priorities for change and growth
- Formulate and evaluate plans for lifestyle modification
- The four major types of intervention by Family Health Nurses in primary, secondary and tertiary prevention, and crisis intervention/direct care.

Qualities of family health nurse:

- Family-centered approach
- Holistic approach (wellness)
- Do not judge during the approach & Self-awareness
- Accept different values and beliefs
- Able to work in adverse conditions
- Flexible
- Sensitive to time and effort
- Independent and positive attitude
- Skilled
- Able to cope with and manage stress
- Able to handle the situation.

School Health services

Introduction

The school health services are important dimension of community. the reasons are large number in the total population, due to rapid growth and development, for early detection of nutritional and communicable diseases, due to group living.

School Health services:

Is defined as "all the strategies, activities, and services offered by, or in association with schools that are designed to promote students physical, emotional, and social development.

School nursing:

Is a specialized practice of professional nursing that advances the well-being, academic success, and facilitate positive student responses to normal human development, promote health and safety in the school environment.

Purpose of School nursing

- Facilitate positive student responses to normal development
- Promote health and safety
- Intervene with actual and potential health problems
- Provide case management services
- Collaborate with others to build student and family for adaptation, self-management, self-advocacy and learning.

Aspects of school health services:

- i. Health Appraisal of school children & School personal.
- ii. Curative measures and follow up.
- iii. Prevention of communicable diseases.
- iv. Healthful school environment.
- v. Nutritional Services.
- vi. First aid and emergency care.

- vii. Mental Health.
- viii. Dental Health.
- ix. Eye health.
- x. Health Education.
- xi. Education of Handicapped children.
- xii. Proper maintenance and use of school health records.

Role of the Community Health Nurse in the school Health:

➤ **The role of the school nurse encompasses both health and educational needs**

- Direct care provider
- Collaborator
- Manager
- Advocate
- Adviser
- Educator
- Consultant

- Planner & organizer
- Implementer
- Supervisor
- Evaluator
- Good observer
- Researcher
- Critical thinker

- Work as a counsellor and educator of health.
- Plan the health talks to be given in school.
- Provide guidance to the teachers and parents in matter of health.
- Work as the Coordinator and organizer of the school health programs.
- Provides direct healthcare to student & staff
- Provides leadership for the provision of health services
- Provides screening and referral for health conditions
- Promotes a healthy school environment
- Promotes health
- leadership role for health policies and programs
- Liaison between school personnel, family, community, and health care providers.
- Help and bridge the gap between what the child learns at school and practices at home.

Elderly Health Service

Introduction:

The Elderly Health Service (EHS) was established to enhance primary health care to elderly people living in the community, improve their self-care ability, encourage healthy living and strengthen family support so as to minimize illness and disability.

Services include:

- **Clinic Service**

Elderly Health Centers aim to address the multiple health needs of the elderly by providing integrated primary health care services to them. Preventive, promotive and curative services are provided from a family medicine perspective using a multi-disciplinary team approach. Elders aged 65 or above are eligible for enrolling as members of Elderly Health Centers. Enrolled members are provided with services of health assessment, counselling, health education and curative treatment.

- **Out-reaching Service**

Visiting Health Teams reach out into the community and residential care settings to provide health promotion activities for the elderly and their caregivers in collaboration with other elderly services providers. **The aim is to** increase their health awareness, the self-care ability of the elderly, and to enhance the quality of caregiving.

- **Public Health and Administration**

The Public Health and Administration Division supports the operation of the Elderly Health Centers and the Visiting Health Teams and provides professional input on elderly health-related issues at an inter-departmental level. Data collected from daily service operations are used for monitoring the health status of the elderly and research purposes.

Students' self-assessment

A. Which activities will the school nurse include when preparing for flu season?

Select all that apply.

- a. Teaching to reuse tissues
- b. Providing hand sanitizers
- c. Instructing on handwashing
- d. Staying at home when febrile
- e. Administering flu vaccinations

B. A home health nurse was making an initial visit to an elderly man. As the nurse began the assessment, the wife gave all the information requested. Which of the following actions would the nurse take next?

- a. Agreeing upon appropriate interventions with the family
- b. Creating appropriate nursing diagnoses
- c. Assessing the environment of the wider community
- d. Confirming the information obtained with the client

C. Discuss the role of community nurse in family health services.

D. Explain the process of referral system.

CHAPTER FOUR

(Environmental and Occupational Health)

- Environment
- Occupational
- Rural health

Environmental Health

Learning objectives

At the end of chapter the student will be able to:-

1. Define the terms *environment*, *ecology*, and *biosphere*.
2. Importance of environmental health for nursing
3. Identify Environmental Principles for Public Health Nursing
4. Identify areas of environmental health, environmental health Problems
5. Enumerate effects of Environmental Hazards
6. Discuss effects of health environment.
7. Discuss nursing role in environmental health

Introduction

Critical theory suggests that nurses must be aware of environmental threats or factors that might detrimentally affect the safety and wellbeing of particular populations or deprive them of access to resources necessary in the pursuit of health. This awareness may include recognizing, supporting, and maintaining positive environmental influences. For instance, a nurse must consider the effects of having access to a safe place to walk on one's ability to maintain healthy.

The **environment** is defined as all the external conditions, circumstances, and influences surrounding and affecting the growth and development of an organism or a community of organisms.

Ecology the study of how living things interact with each other and their environment

Biosphere the zone of the earth where life is found is known as the.

The World Health Organization (WHO) defines **environmental health** as “all the physical, chemical, and biological factors external to a person and all the related factors impacting behaviors.

Importance of Environmental Health for Nursing

- 1- The role of nurses in promoting healing and safe environments for people.
- 2- Nurses work with diverse populations in homes, workplaces, and communities.
- 3- Nurses are one of the most trusted professionals, are able to communicate complex information to their patients and communities, and interact with many other health care organizations.

Principles of Environmental Health for Nursing Practice

1. Knowledge of environmental health concepts is essential to nursing practice.
2. The precautionary principle guides nurses in their practice to use products and practices that do not harm human health or the environment and to take preventive action in the face of uncertainty.
- 3 . Nurses have a right to work in an environment that is safe and healthy.
- 4 . Healthy environments are sustained through multidisciplinary collaboration.
- 5 . Choice of materials, products, technology, and practices in the environment that impact nursing practice are based on the best available evidence.
- 6 . Approaches to promoting a healthy environment reflect a respect for the diverse values, beliefs, cultures, and circumstances of patients and their families.
- 7 . Nurses participate in assessing the quality of the environment in which they practice and live.
- 8 . Nurses, other health care workers, patients, and communities have the right to know relevant and timely information about the potentially harmful products, chemicals, pollutants, and hazards to which they are exposed.
- 9 . Nurses participate in research of best practices that promote a safe and healthy environment.
10. Nurses must be supported in advocating for and implementing environmental health principles in nursing practice.

AREAS OF ENVIRONMENTAL HEALTH

AREA	DEFINITION	PROBLEMS
Built environment	Buildings, spaces, and products that are created or modified by people, including homes, schools, workplaces, parks/recreation areas, greenways, business areas, and transportation systems.	- Drunk driving - Secondhand smoke - Noise exposure - Urban crowding - Technological hazards
Work related exposure	Occupational exposure to environmental hazards that can cause illness or injury.	- Asbestos exposure - Agricultural accidents - Excessive exposure to x-rays
Outdoor air quality	The protectiveness of the atmospheric layers, the risks of severe weather, and the purity of the air for breathing purposes.	- Gaseous pollutants - Greenhouse effect - Destruction of the ozone layer - Aerial spraying of herbicides and pesticides - Acid rain - Nuclear facility emissions
Healthy home	The availability, safety, structural strength, cleanliness, and location of shelter, including public facilities and family dwellings. This includes indoor air quality.	- Homelessness - Rodent and insect infestation - Presence of lead-based paint - Sick building syndrome - Unsafe neighborhoods - Radon gas seepage in homes and schools

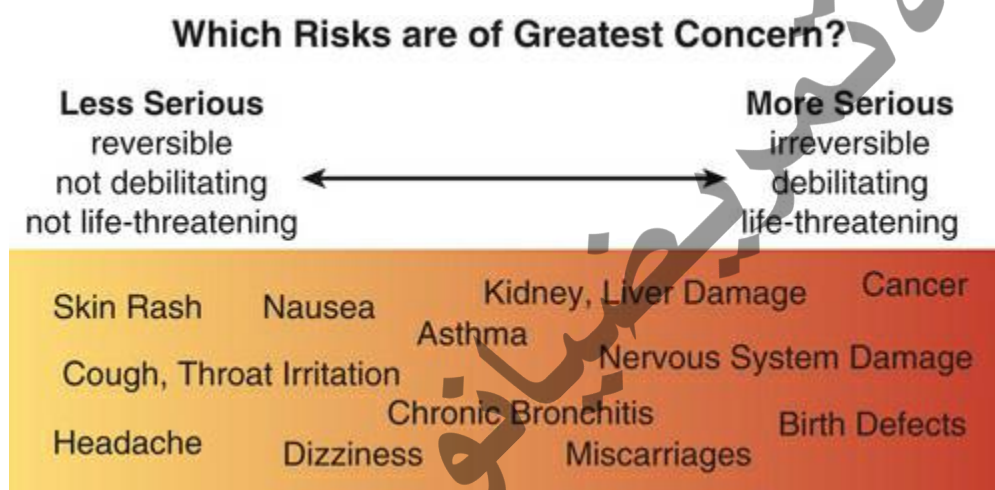
Water Quality	The availability of and accessibility to a clean water supply, the mineral content levels, pollution by toxic chemicals, and the presence of pathogenic microorganisms.	- Contamination of drinking supply by human waste - Oil spills in the world's waterways - Pesticide or herbicide infiltration of groundwater - Aquifer contamination by industrial pollutants - Heavy metal poisoning of fish
Food safety	The availability, relative costs, variety, safety, and health of animal and plant food sources.	- Malnutrition - Bacterial food poisoning - Food adulteration - Disruption of food chains by ecosystem destruction - Carcinogenic chemical food additives
Waste management	The management of waste materials resulting from industrial and municipal processes, human consumption, and efforts to minimize waste production.	- Use of nonbiodegradable plastics - Poorly designed solid-waste dumps - Inadequate sewage systems - Transport and storage of hazardous waste - Illegal industrial dumping - Radioactive hazardous wastes

Effects of Environmental Hazards

Effects of environmental hazards may be general or specific.

- **General effects:-** The ramifications of high unemployment, drought, and extensive smog cover affect the public generally.
- **Specific effects:-** The housing needs of elderly people who use walkers or canes, the occupational risks of electrical line repair workers, and the mentally incapacitating effects of elevated blood lead values in children, affect the public more specifically.

Degrees of health effects from environmental exposure



Environmental health effects

Environmental health effects can be immediate, long-term, or Inter-generational.

Immediate environmental health effect

Burns, gunshot wounds, hurricane damage, and outbreaks of gastrointestinal distress among cafeteria customers.

Long-term environmental health effect

Gradual occupational hearing loss, “black lung” in coal miners, and increased rates of thyroid cancer among young victims of the Chernobyl nuclear reactor accident.

Intergenerational environmental health effects

Will likely occur with climate change by affecting women of childbearing age.

Example of effort to monitoring and improving environmental health is annually UN climate change conference.

Every year. Countries who have joined the united nation framework conversation on climate change (UNFCCC) meet to: -

- Measure progress and negotiate multilateral response to climate change.
- Stabilize use gas concentration in the atmosphere at a level that will prevent dangerous human interference with the climate system

Nursing Role in Environmental Health

1. Nurses must work with the public to promote more stringent and
2. Actively enforced environmental legislation and regulations.
3. Actions must include not only national but also worldwide environmental policies.
4. Environmental concerns for clean air, clean water, and freedom
5. From noxious chemicals must become nursing concerns.
6. Community health nurses can be catalysts to neighborhood efforts to produce safe living environments.
7. Community health nursing must expand its theory and practice to incorporate the fact that individual and community health ultimately depends on global environmental integrity.
8. Nursing must include an environmental perspective by committing to environmental health promotion initiatives that promote social justice and environmental responsibility.

OCCUPATIONAL HEALTH NURSING

Learning objectives

At the end of chapter the student will be able to:-

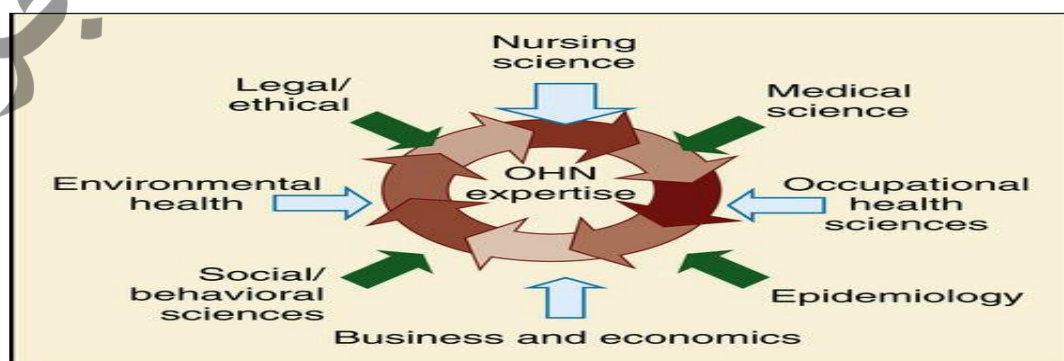
- 1- Define occupational health.
- 2- Identify occupational health nursing services.
- 3- Enumerate priorities in occupational health nursing.
- 4- Discuss types of occupational hazards.
- 5- Demonstrate skills of occupational health nursing.

Introduction

Occupational health, including **toxicology**, to recognize routes of exposure, examine relationships between chemical exposures in the workplace and acute and latent health effects such as burns or cancer, and understand dose-response relationships; **industrial hygiene**, to identify and evaluate workplace hazards so control mechanisms can be implemented for exposure reduction; safety, to identify and control workplace injuries through active safeguards and worker training and education programs about job **safety**; and **ergonomics**, to match the job to the worker, emphasizing capabilities and minimizing limitations.

A subspecialty of public health nursing is defined by the American Association of Occupational Health Nurses (AAOHN) as the following:

- The specialty practice that focuses on the promotion, prevention, and restoration of health within the context of a safe and healthy environment.
- It includes the prevention of adverse health effects from occupational and environmental hazards.
- It provides for and delivers occupational and environmental health and safety programs and services to clients.



Occupational health nursing knowledge domains

Occupational Health Nursing Services

- Safe and healthful workplace
- Emergency medical response
- First aid responder selection and training
- First aid space, supplies, protocols, and records
- Designated medical resources for incident response
- Workers' compensation
- Confidentiality of medical records
- Compliance with medical record retention requirements
- Occupational Safety and Health Administration (OSHA) compliance
- Medical personnel requirement
- Injury and illness reporting and recording
- Accident and injury investigation
- Cumulative trauma disorder prevention
- Employee access to medical and exposure records
- Medical surveillance and hazardous work qualification
- Personal protective equipment evaluation and training
- Infection control
- Employee Right-to-Know Act notification and training
- Community Right-to-Know Act compliance
- Rehabilitation Act: handicap, preplacement, fitness for duty evaluations, accommodations
- Department of Defense, Department of Transportation, Nuclear Regulatory Commission, and Drug-Free Workplace Act compliance
- Policy development
- Drug awareness education
- Drug testing and technical support
- Employee Assistance Program services
- Threat of violence and duty to warn
- State and local public health regulations
- Nursing practice acts
- Board of Pharmacy and Drug Enforcement Agency regulations

- Continuing professional education required for licensure Services Often Mandated by Company Policy
- Clinical supervision of on-site health services
- Health strategy development
- Health services standards
- Space, staffing, and operational standards
- Occupational illness and injury assessment, diagnosis, treatment, and referral
- Nonoccupational illness and injury assessment, diagnosis, treatment, and referral
- Disability and return-to-work evaluations and accommodations
- Impaired employee fitness for duty evaluation
- Preplacement evaluation and medical accommodation
- Handicap evaluation, placement, and accommodation
- Employee Assistance Program standards
- International health: travel, medical advisory, and immunizations
- Data collection and analysis
- Medical consultation
- Pregnancy placement in hazardous environments
- Professional education and development
- Audit and quality assurance

Priorities In Occupational Health Nursing

- Effectiveness of primary health care delivery at the worksite
- Effectiveness of health promotion nursing intervention strategies
- Nature and effects of stress and workplace stressors on worker health
- Strategies that minimize work-related adverse health outcomes (e.g., respiratory disease)
- Health effects resulting from chemical exposures in the workplace
- Occupational hazards of health care workers (e.g., latex allergy and blood-borne pathogens)
- Factors that influence workers' rehabilitation and return to work
- Effectiveness of ergonomic strategies to reduce worker injury and illness
- Health effects resulting from the interaction between aging and workplace hazards

- Evaluation of critical pathways to effectively improve worker
- Health and safety and enhance maximum recovery and safe return to work
- Evaluation of intervention strategies to improve worker health and safety
- Strategies for increasing compliance with or motivating workers to use personal protective equipment
- Emergency/pandemic preparedness in the workplace
- Impact of occupational health nursing interventions on worker's compensation claims

Occupational hazards and associated health effects



Types of occupational hazards and associated health effects

CATEGORY	EXPOSURES	HEALTH EFFECTS
Biological	Blood or body fluids	Bacterial, fungal, and viral infections (e.g., hepatitis B)
Chemical	Solvents	Headache and central nervous system dysfunction
	Lead	Central nervous system disturbances
	Asbestos	Asbestosis
	Acids	Burns
	Glycol ethers	Reproductive problems

	Mercury	Ataxia
	Arsenic	Peripheral neuropathy
Enviromechanical	Static or non-natural postures	Musculoskeletal disorders
	Repetitive or forceful exertions	Back injuries
	Lighting	Headache and eye strain
	Shift work	Sleep disorders
	Electrical	Electrocution
	Slips and falls	Musculoskeletal conditions
	Struck by or against object	Injury
Physical	Noise	Hearing loss
	Radiation	Reproductive effects and cancer
	Vibration	Raynaud's disease
	Heat	Heat exhaustion and heat stroke
Psychosocial	Stress	Anxiety reactions and a variety of physical symptoms

Skills of Occupational Health Nursing

Clinical and Primary Care

- Applying the nursing process in delivery of care.
- Providing first aid and primary care according to treatment protocols.
- Conducting a physical assessment.
- Taking an occupational and environmental health history.
- Diagnosing and treating.
- Being knowledgeable about immunization protocols.
- Identifying employees' emotional needs and providing support and counseling.
- Using a multidisciplinary problem-solving approach to occupational health illness and injury.
- Maintaining records.
- Clinical testing and monitoring.
- Responding to medical emergencies.
- Being knowledgeable about trends in health-related issues.

Case Management

- Identifying the need for case management services
- Conducting case management assessments using a multidisciplinary framework
- Developing case management care plans

- Evaluating resources and vendors for case management
- Implementing early return-to-work programs
- Monitoring and evaluating outcomes
- Developing policies and programs for case management
- Analyzing trends for case management services
- Designing disability management systems
- Conducting case management outcomes-based research

من فضاء نهج ريادة الأعمال على التليجرام

RURAL HEALTH

Learning objectives

At the end of chapter the student will be able to:-

- 1- Define rural health.
- 2- Illustrate the rural health disparities
- 3- Discuss causes of rural health disparities.
- 4- Discuss role of community health nursing in rural health.

Introduction:

Rural nursing practice offers many opportunities. Nurses are respected community members—their judgment and opinions count. Rural nurses are key members of the health care team. They can make a difference in the lives of their neighbors, friends, and community. Rural public health nurses (PHNs) often struggle with helping clients gain access to quality health care and the inherent transportation problems found in isolated areas. The challenges are many, but the rewards are great.

Definitions of rural health:

Rural health is the health of people living in rural areas, who generally are located farther from health care facilities and other services than people living in urban areas.

The rural health disparities

- Rural residents are at greater risk of death from:-
 - Heart disease
 - Cancer
 - Stroke
 - Unintentional injury (for example: motor vehicle death)(
 - Chronic lower respiratory disease
- Rural residents also are more likely to have multiple health conditions compared to people living in other areas.
- Heart disease, stroke, hypertension, diabetes, and arthritis are examples of health conditions that are more common in rural areas.
- Within rural areas, members of racial/ethnic minority, tribal, and other diverse groups are at higher risk for poor health outcomes

Causes rural health disparities

Factors that put people living in rural areas at greater risk for poor health outcomes include:

1. Less access to care because of fewer health care providers, medical facilities, and transportation options.
2. Exposures to specific environmental hazards, such as pollution and poor air and water quality
3. Higher rates of cigarette smoking, high blood pressure, and obesity
4. Less leisure-time physical activity
5. Lower seatbelt use
6. Food deserts, areas with fewer grocery options and lower availability of healthy foods
7. Higher rates of poverty and less likelihood of having health insurance

Nursing role in rural community health.

1. **Advocate:** Assists rural clients and families in obtaining the best possible care
2. **Coordinator/case manager:** Connects rural clients with needed health and social services, often assisting with information on transportation
3. **Health teacher:** Provides education to individuals, families, or groups on health promotion or other health-related topics (e.g., prepared childbirth, parenting, diabetes maintenance, home safety)
4. **Referral agent:** Makes appropriate connections between rural clients and urban service providers
5. **Mentor:** Guides new community health nurses, nursing students, and other nurses new to the rural community
6. **Change agent/researcher:** Suggests new approaches to solving patient care or community health problems based on research, professional literature, and community assessment
7. **Collaborator:** Seeks ways to work with other health and social service professionals to maximize outcomes for individual clients and the community at large
8. **Activist:** With a deep understanding of the community and its population, takes appropriate risks to improve the community's health.

CHAPTER FIVE

Communicable and non-communicable diseases

1-COMMUNICABLE DISEASES

Learning objectives

At the end of chapter the student will be able to:-

- 1- Define communicable diseases
- 2- Identify the basic terminology
- 3- Illustrate chain of infection
- 4- Determine risk factors of communicable diseases
- 5- Mention Examples of common communicable
- 6- Determine Prevention of communicable diseases
- 7- Mention nursing role in caring of patients with communicable diseases

Introduction

Communicable diseases occur in every country, in every urban and rural area, and in every neighborhood, from the very rich to the very poor. Nurses who provide quality care in combating communicable diseases must have a basic understanding of epidemiology, infection control, microbiology, medicine, public health, and nursing. Furthermore, the community nurse must have knowledge of the legal system, which mandates prevention and control of communicable diseases locally, nationally, and worldwide. Nurses must also have knowledge of effective support systems that can be used by individuals, families, and communities.

Definition of communicable diseases

Is one that can be transmitted from one person to another and is caused by an infectious agent that is transmitted from a source or reservoir to a susceptible host.

Basic Terminology

- **Infection**: is the entry and development or multiplication of an infectious agent in the body of man or animals. An infection does not always cause illness
- **Endemic**: Refers to the constant presence of a disease within a geographic area or a population. (e.g) pertussis is endemic in us.
- **Epidemic**:- Refers to the occurrence of disease in a community or region in excess of normal expectancy.
- **Pandemic**: Refers to an epidemic occurring worldwide and affecting large populations. e.g. Swine flu
- **Incubation period** : The time interval between exposure to an infectious agent and the appearance of the first sign or symptom of the disease in question.
- **Elimination**:- Its goal is to remove a disease from a large geographical area such as a country or region of the world ,(through programs and campaign) .(e.g) elimination of (polio) from the America 1994.
- **Eradication**:- Its goal is to get rid of the disease completely by extermination of the infectious agents worldwide ,(through programs and campaign) (e.g) eradication of smallpox on May 1980

Chain of infection

- **Infectious agent** :any disease causing microorganism (pathogen), Infectious agents are : Bacterial ,Viral ,Fungal , Parasitic
- **The reservoir**: Where a microorganism normally lives and reproduces , it may be Humans(case or carrier),Animals ,Water , Food
- **Portal of exit**: route of escape of the pathogen from the reservoir, as respiratory secretions, blood, breaks in skin, Ex: Flu or cold through mucous secretions, Hepatitis A through stool, T. B through droplet
- **Mode of transmission**

1-Direct

- A-direct contact (touching/ cough)
- B-air born (droplet-dust –droplet nuclei)
- C- trans-placental

2-In direct

A-vehicle born (food/blood/toys/ fumets)

B-vector born (mechanical)

C-air born

- **The Portal of Enter** : Route through which the pathogen enters its new host , through inhalation , ingestion , and sexual contact, breaks in protective skin barriers.
- **The Susceptible Host**: The organism that accepts the pathogen, may be man or animal.

Risk factors of communicable

1. **Age**: children under five years , orders are risky
2. **Occupation**: some occupation are more risky to infection as farmers who are in contact with animals, medical and paramedical personnel.
3. **Sex** : some infections are more in males and some are more in females
4. **Social factors**: low socioeconomic class, poverty, bad health habits
5. **Environmental factors**: poor sanitary houses, overcrowding.
6. **Malnutrition**: leads to low resistance
7. **Chronic and debilitating disease**: condition that lower the resistance as diabetes mellitus, HIV
8. **Immunity**: low immunity can be risk factor
9. **Psychological factor**

Examples of common communicable

1- Human immunodeficiency virus (HIV)

- ❑ HIV (human immunodeficiency virus) is a virus that attacks cells that help the body fight infection, making a person more vulnerable to other infections and diseases.

Mode of transmission

- Infected fluids must be exposed to damaged tissue or areas of the body containing mucosal membranes, such as the rectal or vaginal lining, to spread the virus.
- Mothers can transmit the virus to infants during pregnancy or childbirth

- Injection from a contaminated needle or syringe may also transmit the virus directly into the bloodstream.

Signs and symptoms

- Fever.
- Headache.
- Muscle aches and joint pain.
- Rash.
- Sore throat and painful mouth sores.
- Swollen lymph glands, mainly on the neck.
- Diarrhea.
- Weight loss.

Prevention

- reduces the incidence of transmission (e.g., fewer people become HIV infected)
- Avoid sexual contact with infected individuals
- Avoid sharing needles/syringes that could be contaminated with HIV
- Avoid any type of contact with the bodily fluid of an infected individual.

2-Hepatitis C

- Hepatitis C is a liver infection that can lead to serious liver damage .
- It's caused by the hepatitis C virus. But it causes few symptoms, so most of patients don't know .

Mode of transmission

- The virus spreads through an infected person's blood or body fluids.
- Sharing injection drugs and needles
- Having sex, especially if you have an STD, an HIV infection, several partners, or have rough sex
- Being stuck by infected needles
- Birth -- a mother can pass it to a child
- Sharing personal care items like toothbrushes, razor blades, and nail clippers.

Signs and symptoms

- Clay-colored poop
- Dark urine
- Fever
- Fatigue
- Jaundice (a condition that causes yellow eyes and skin, as well as dark urine)
- Joint pain
- Loss of appetite
- Nausea
- Stomach pain

Prevention

A-Health education: to educate the public about mode of transmission and methods of protection

- Avoid sharing personal items such as razors or tooth brushes,
- Finger tools.
- Do not share drug needles
- Clean blood spills

B-Health promotion:

Good nutrition, healthy life style.

- Eat whole-grain cereals, breads, and grains
- Eat lots of fruits and vegetables
- Get enough protein
- Drink plenty of fluids

3- Pulmonary Tuberculosis

- **Infectious** disease which may involve any part of the body, pulmonary T.B. is the most frequent.

Modes of transmission:

- **Direct droplet:** organism find exit through respiratory secretions on coughing.
- **Indirect:** airborne, toy

Signs and symptoms

- Lack of appetite and weight loss
- A high temperature
- Night sweats
- Extreme tiredness or fatigue
- A persistent cough that lasts more than 3 weeks and usually brings up phlegm, which may be bloody
- Breathlessness that gradually gets worse

Prevention:

General prevention:

- Sanitary environment
- **Health education:** to educate the public about mode of transmission and methods of prevention
- **Health promotion:** good nutrition, healthy life style.

Specific prevention:

- **Vaccination:** By BCG Administered to infants, Dose:1 ml intradermal

4- Typhoid

Typhoid fever is a food borne infection caused by the consumption of food or water contaminated with the bacterium Salmonella enterica

Mode of transmission:

Transmission from host to host occurs if food contaminated with such fecal matter is ingested (the fecal-oral pathway)

Signs and symptoms

- Malaise and coated tongue
- Headache and cough.
- Abdominal pain is also possible.
- Rose spots may be detected
- abdominal distended with constipation ,more than Diarrhea in adult

- Diarrhea can occur in this stage: six to eight stools in a day, green color with pea –soup odor.
- The spleen and liver are enlarged (hepatosplenomegaly) and tender.

Prevention

- Avoid contaminated food.
- Hand washing especially before food preparation and eating
- Take typhoid vaccination

5-Covid 19

Corona viruses are a family of viruses that can cause illnesses such as the common cold, severe acute respiratory syndrome (SARS) and middle east respiratory syndrome (MERS) In 2019, a new corona virus was identified as the cause of a disease outbreak that originated in Wuhan, China , in December 2019.

Mode of Transmission:

- Infected people exhale those particles as they breathe, talk, cough, sneeze, or sing
- Inhaled or land on mucous membranes on the eyes, nose, or mouth to infect.
- Touching a contaminated surface or object before touching their own mouth, nose, or Eyes.

Signs and Symptoms:

- Pain
- Headaches
- Fatigue
- Shortness of breath
- Anxiety
- Depression
- Fever
- Chronic cough
- Sleep problem

Prevention

- Wear a mask that covers your nose and mouth. Make sure that your hands are clean when you put on and remove your mask.
- Keep a physical distance of at least 1 meter from others.
- Avoid poorly ventilated or crowded spaces.
- Open windows to improve ventilation indoors.
- Wash your hands regularly.
- When it's your turn, get vaccinated.
- Regularly clean your hands often. Use soap and water alcohol-based.
- Get fully vaccinated or boost your immunity.

Nursing role in preventing communicable diseases

- 1- Communicable disease prevention and control programs must move beyond providing drugs and treatment.
- 2- Monitor and provides case management services to individuals with communicable disease
- 3- Health education and promotion must be aimed at changing individual and community behavior
- 4- Collects, analyzes and report communicable disease and epidemiological data
- 5- Conducts communicable disease, epidemiological, and other studies under direction and collaboration with the health officer and communicable disease manager
- 6- Act as resource and provide consultation and health information to individuals, families.
- 7- Mobilizing community participation is essential to successful implementation of the program.

2-Noncommunicable diseases

Learning objectives

At the end of chapter the student will be able to:-

- 1- Define non-communicable diseases
- 2- State Risk factors of non-communicable diseases
- 3- Mention Examples of common noncommunicable diseases
- 4- Determine Prevention of noncommunicable diseases
- 5- Mention Nursing role in caring of patients with communicable diseases

Definition of noncommunicable diseases

Chronic conditions that do not result from an infectious process and not communicable

Risk factors of noncommunicable diseases

- Heredity.
- Obesity.
- Sleep apnea.
- Advancing age.
- Ethnicity/race.
- Increased sodium intake.
- Diabetes mellitus.
- Smoking.
- Lack of physical exercise.

Examples of common communicable

1- Ischemic heart disease

Definition of ischemic heart disease

- Ischemic heart disease is the obstruction of blood flow
- Through the coronary arteries to the heart muscle cells, typically
- From atherosclerosis. Blood flow reduction resulting
- From IHD can cause angina, MI, or sudden death if blood
- Flow is not restored.

Types of ischemic heart disease

1- Angina Pectoris.

2- ACUTE CORONARY SYNDROME.

- Unstable angina.
- Myocardial infarction (MI).

A. Angina pectoris

Stable angina

This is the most common type of angina, and pain is usually brought about by physical exertion that results in the heart rate increasing. This type of angina is defined as stable because there is a regular pattern for the pain and similar triggers exist for each attack of angina..

VARIANT OR VASOSPASTIC ANGINA (PRINZMETAL'S ANGINA).

This type of angina is caused by coronary artery spasms and is serious. The pattern of occurrence is often cyclical, with the pain presenting, about the same time each day.

B. Acute coronary syndrome

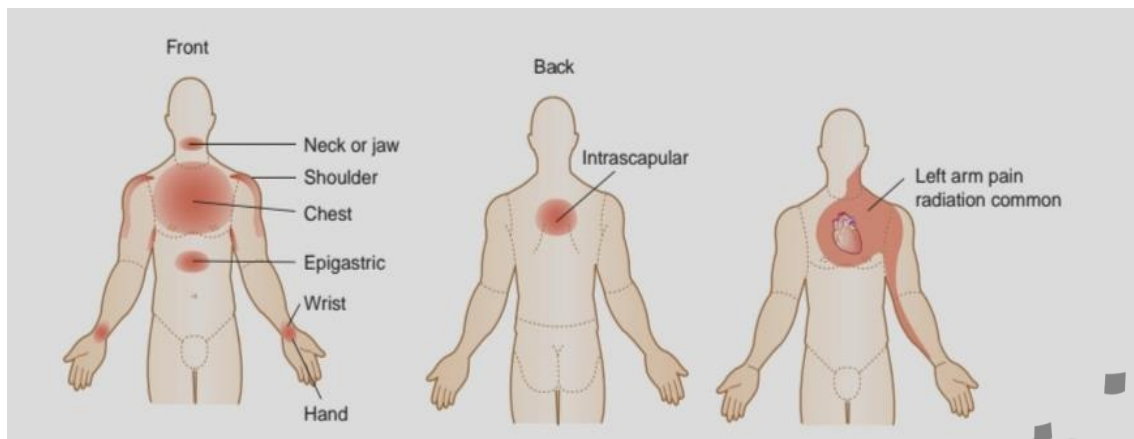
• Unstable angina

Unstable angina does not follow a regular pattern. A small amount of physical effort such as walking, or even just resting, can result in an attack. In many cases, stable angina can progress to unstable angina, which may be an indication of worsening of coronary heart disease.

• Myocardial infarction

Signs and symptoms of ischemic heart disease

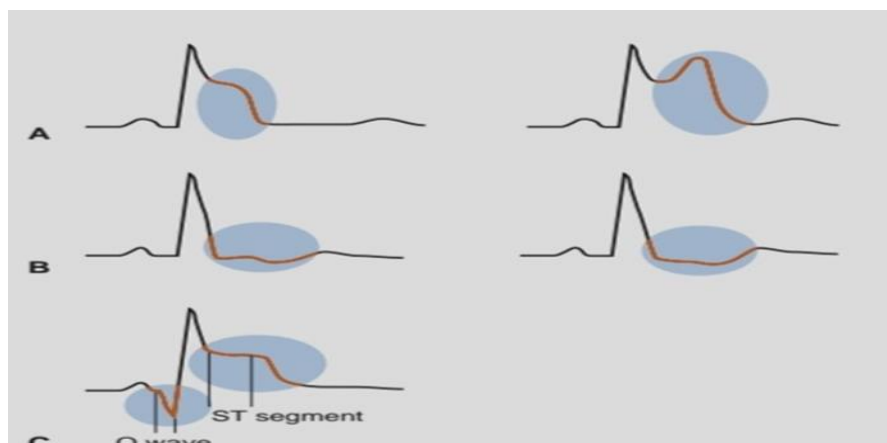
- 1- Patients (especially men) often describe the pain as heaviness, tightness, squeezing, viselike, or crushing in the center of the chest or adjacent to the chest.
- 2- The pain can radiate down one or both arms, with pain in the left arm being more common, into the shoulder, neck, jaw, or back.



- 3- Patients may also describe heaviness in their arms or a feeling of impending doom.
- 4- During the episode of pain, the patient may be pale, diaphoretic, or dyspneic.
- 5- The pain is usually brought on by exertion and subsides with rest.
- 6- It can be relieved with a vasodilator such as NTG.
- 7- Episodes of chest pain may increase in frequency and severity over time. If patients do not heed the warning to stop their activity and rest, they may be at risk for MI or sudden death.
- 8- Consider the elderly in whom because of neuropathy, the chest pain of an MI may not be present.
- 9- In women, the MI symptomatology may also include abdominal discomfort, fatigue, heartburn, or back pain, with fatigue being the greatest indicator.
- 10- Nausea, vomiting.
- 11- Changes in the level of consciousness (LOC) without other causes.
- 12- Syncope or near syncope episodes.

Diagnostic evaluation for ischemic heart disease

- 1- ECG.



- 2- Serum cardiac troponin I or T.
- 3- Serum myoglobin.
- 4- Serum CK-MB.
- 5- CBC.
- 6- Serum magnesium and potassium.
- 7- Vital signs, oxygen saturation, intake and output.
- 8- Echocardiogram.
- 9- Possible coronary interventions: percutaneous transluminal coronary angioplasty (PTCA), angioplasty, stents, atherectomy, left ventricular assist devices, or cardiac bypass.

Management for patient with ischemic heart disease

Medical management

- 1- Oxygen.
- 2- Morphine sulfate.
- 3- Nitrates.

Vasodilate to reduce preload and afterload. Reduce oxygen consumption of myocardium.

- 4- Fab Four cardiac medications: aspirin, statin, ACEI, beta Blocker.

- **Angiotensin-Converting Enzyme Inhibitors**

Block production of angiotensin II, a potent vasoconstrictor. Vasodilate and improve cardiac output and exercise tolerance.

- **Calcium Channel Blockers**

Dilate peripheral arteries, decrease myocardial contractility, depress conduction system, and decrease workload of the heart. In variant angina, reduce coronary artery spasm.

- **Beta Blockers**

Decrease pulse, BP, and cardiac output and suppress renin activity. Decrease the risk of sudden death.

- 5- Platelet aggregation inhibitors.

- Inhibit platelet activation, adhesion, or procoagulant activity.

- 6- Thrombolytics.

- 7- Anticoagulants.

- 8- Antidysrhythmic.
- 9- Vasodilators.
- 10- Percutaneous coronary interventions and stents.

2- Diabetes Mellitus

Definition of diabetes mellitus

Diabetes is a group of metabolic diseases characterized by hyperglycemia resulting from defects in insulin secretion, insulin action, or both.

Classification of diabetes

(WHO) Expert Committee on Diabetes

Mellitus which classified diabetes based on the age of onset of the disease, mainly:

- 1- Juvenile-onset diabetes (type 1)
- 2- Maturity-onset diabetes (type 2)
- 3- Gestational diabetes (GDM)

Signs and symptoms of diabetes

- 1- Polyurea
- 2- Polyphagia
- 3- Polydipsia

Methods for diagnosing

Diabetes

- 1- An HbA1c of $\geq 6.5\%$
- 2- Fasting blood glucose
- 3- Two-hour plasma glucose during an oral glucose tolerance testing (OGTT)
- 4- Random plasma glucose

Treatment of Diabetes

1- Medical Nutrition Therapy

Maintain blood glucose and lipid levels as near to normal as possible to prevent long-term complications. For some, especially those with type 2 diabetes, weight loss

and cholesterol and blood pressure control may be additional goals of nutrition therapy.

2- Exercise

- Exercise lowers blood glucose by improving the sensitivity of muscles to insulin, and the effects can last up to 48 hours.
- Exercise also improves blood lipid levels and circulation, which is important for the person with diabetes who already has an increased risk of cardiovascular disease

3- Medication

- a- INJECTED INSULIN.** The person with type 1 diabetes has no endogenous insulin and therefore must inject insulin daily.
- b- ORAL HYPOGLYCEMIC MEDICATION.** The patient with type 2 diabetes may be able to control blood glucose levels with medical nutrition therapy and exercise alone.

3- Cancer

Definition of Cancer:

Cancer is a group of cells that grows out of control, taking over the function of the affected organ.

Cancer Classification:

- 1- *Carcinoma* is the most common type of cancer and includes cells of the skin, gastrointestinal (GI) system
- 2- *Sarcomas* is cancer cells affecting connective tissue, including fat, the sheath that contains nerves, cartilage, muscle, and bone.
- 3- *Leukemia* is the term used to describe the abnormal growth of white blood cells (WBCs).
- 4- *Lymphomas* cancers involving cells of the lymphatic system, lymph nodes, and spleen.

Diagnosis of Cancer:

- 1- Biopsy.
- 2- Radiological procedures.
- 3- Nuclear imaging procedures.
- 4- Ultrasound procedures.
- 5- Magnetic resonance imaging.
- 6- Endoscopic procedures.
- 7- Laboratory tests.
- 8- Cytological study.

TREATMENT FOR CANCER

1- SURGERY.

Surgery can be curative when it is possible to remove the entire tumor. Skin cancers and well-defined tumors without metastasis can be removed without any additional intervention. For some tumors, as much of the tumor is removed as possible (this is called debulking), and follow-up chemotherapy or radiation is used to treat the remaining tumor cells.

2- RADIATION.

Radiation is used commonly in cancer control and palliation, and it can be curative if the disease is localized. The decision to use radiation is commonly based on cancer site and size. Radiation destroys cancer cells by affecting cell structure and the cell environment. It is used in fractionated (divided) doses to prevent destructive side effects

3- CHEMOTHERAPY.

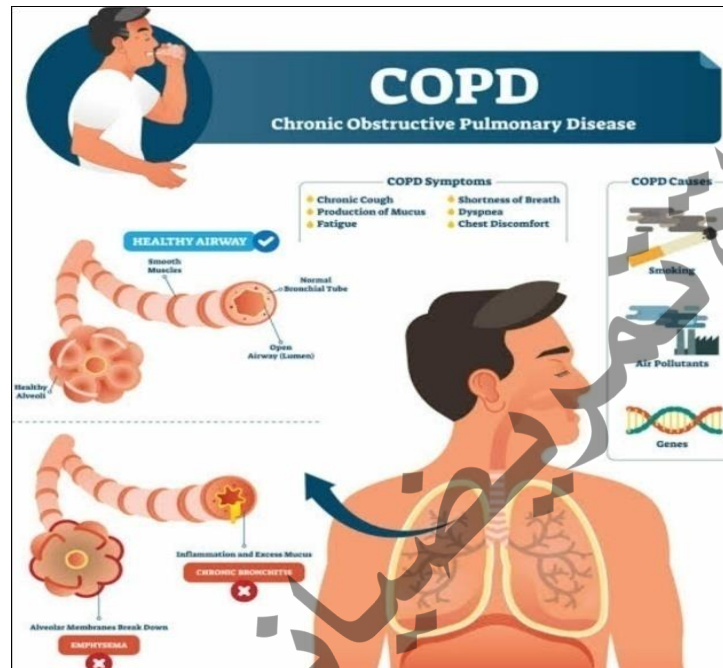
Is chemical therapy that uses **cytotoxic** drugs to treat cancer, cytotoxic drugs can be used for cure, control, or palliation of cancerous tumors. They are classified according to how they affect cell activity.

4- CHRONIC OBSTRUCTIVE PULMONARY DISEASES (COPD)

Definition

Is the name for a group of lung condition that cause breathing difficulties such as (emphysema, chronic bronchitis)

- **Emphysema:** damage to the air sacs in the lungs
- **Chronic bronchitis:** long term inflammation of the airway



Signs and symptoms:

- 1- Breathlessness specially when active
- 2- Persistent chesty cough with phlegm
- 3- Frequent chest infection
- 4- Persistent wheezing

Treatment of COPD:

- 1- Inhalers and medicines to help make breathing easier
- 2- Pulmonary rehabilitation – a specialized programme of exercise and education
- 3- Stop smoking
- 4- Surgery or lung transplantation

Prevention of non-communicable diseases

Follow a healthy life style as:

a- Nutrition

- Eat food that contain all nutrients
- Avoid fatty food
- Avoid adding too much sugar or salt in your food
- Avoid too much carbohydrates in your food specially for diabetic patient.
- Eat small frequent meals
- Drinking plenty of water at least 2 liter per day

b- Exercise regularly at least walking daily for 30 min

c- Avoid stress

d- Quit smoking

e- Avoid alcohol

f- Loss weight

[

Nursing role in caring of patients with non-communicable diseases

- 1- Prevention, treatment and management
- 2- Detect, treat and refer patients
- 3- Point of the first contact
- 4- Provide behavioral and lifestyle interventions
- 5- Monitor blood pressure and blood glucose level regularly.
- 6- Educate patient how to follow healthy life style
- 7- Instruct patient to follow drug regimen according to doctor order
- 8- Educate patient about possible symptoms that require visiting doctor as chest pain, severe headache, hyperglycemia, facial drop, unable to move, dyspnea.
- 9- Encourage patients to increase their literacy about the disease
- 10- Advocate for health policies that integrate noncommunicable disease prevention into health planning.

Vulnerable population and care of them

Learning objectives

At the end of chapter the student will be able to:-

- 1- Define vulnerable population**
- 2- State Categories of vulnerable population**
- 3- List Principles for caring of vulnerable population**
- 4- Mention Care of vulnerable population**

Introduction:

Although the word 'vulnerable' is now widely used in health and social care, its precise definition remains elusive. This is partly because of its universality and dynamism in that everyone becomes or feels vulnerable to greater or lesser extent at different times in their lives. There are also many other reasons. These include the fact that it has a variety of meanings, such as that a person is in danger, at risk under threat, susceptible to problems, helpless, and in need of protection and/support

Definition of vulnerable population

Those groups with increased risk for developing adverse health outcomes.

Categories of vulnerable population

1. Loneparents.

Is usually defined as a divorced, separated, single or widowed mother or father living without a spouse (and not cohabiting) with his or her never-married dependent child or children.

People with disabilities:

The term 'disability' will be used in its broadest sense and will refer to physical and mental disabilities.

2. Older people.

Some older people lead independent and active lives, participating in, and contributing to their families, communities and society in numerous ways. However, others are among the most socially excluded and vulnerable in Society.

3. Children.

Childhood was not a distinct stage of life in most cultures in that children were not regarded as significantly different from adults. For instance, they were integrated into adult working and social life, treated and dressed as miniature adults who gradually assumed adult roles and slowly matured into adulthood.

4. The mentally ill.

Many people have real psychiatric problems but never receive any help.

5. The homeless.

These include the fact that they do not feel emotionally secure and safe in their home, they do not have any rights to stay there and if their accommodation is substandard and adversely affects their health. Many people assume that homelessness means sleeping on the streets. It is a far broader concept in that a person or a family can be homeless for a variety of different reasons.

Principles of caring for vulnerable population:

- 1- Empowerment.
- 2- Protection.
- 3- Prevention.
- 4- Proportionality.
- 5- Partnership.
- 6- Accountability.

Care of vulnerable population:

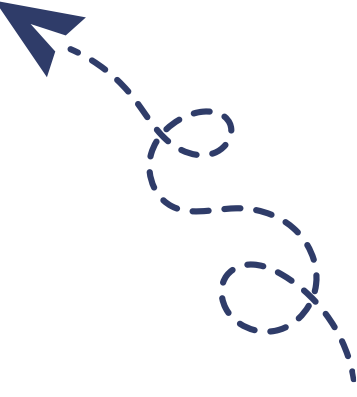
- 1- recognize the vulnerable population
- 2- screen the patient
- 3- provide obligatory immunization to children by the ministry of health and use law to those who miss immunization dose
- 4- Public health facilities provide maternal and child health services.
- 5- Provide financial support to lone parents by the ministry of welfare
- 6- Provide with assisting devices to those with disability as walking aids, wheel chairs and cans.

- 7- Provide home to homeless to eradicate street children phenomena
- 8- Improve social determinates to promote healthy living
- 9- Utilize a global budgeting national healthcare system
- 10- Provide access to virtual healthcare
- 11- Match hospitalization needs to surrounding communities
- 12- Support community-appropriate healthcare access

من فضاء نهضة مصر بـ"مركز دراسات
مصر" على التليجرام

References

- Gangadharan, S., & Hemamalini, M. (2021). *Community Health Nursing: Framework for Practice: Vol 2-E-Book*. Elsevier Health Sciences.
- Rector, C., & Stanley, M. J. (2020). *Community and public health nursing*. Lippincott Williams & Wilkins.
- Schoon, P. M., & Porta, C. M. (2023). *Population-Based Public Health Clinical Manual: Using EBP to Transform Policy in Nursing and Healthcare*. Sigma Theta Tau.
- Sharma, M., & Branscum, P. W. (2020). *Introduction to community and public health*. John Wiley & Sons.
- Stanhope, M., & Lancaster, J. (2021). *Foundations for population health in community/public health nursing-E-book*. Elsevier Health Sciences.
- Chilton, S., & Bain, H. (2025). *A textbook of community nursing*. Third edition. Routledge



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